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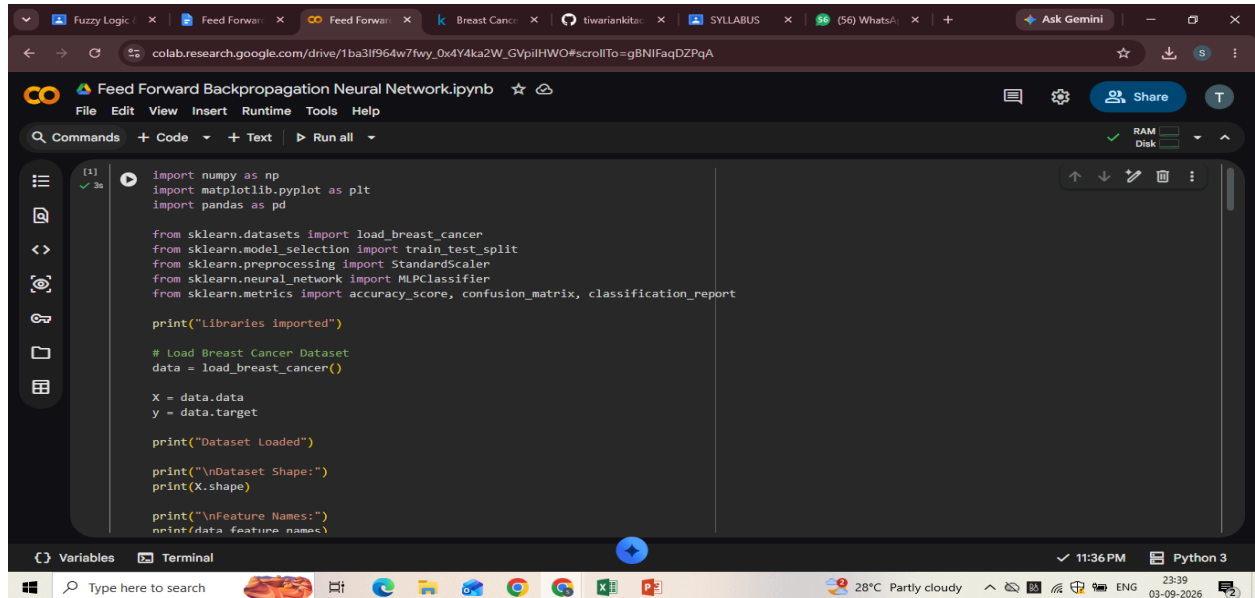
AIM: Feed Forward Backpropagation Neural Network

Implement the Feed Forward Backpropagation algorithm to train a neural network.

Use a given dataset to train the neural network for a specific task.

Evaluate the performance of the trained network on test data.

INPUT:



```
[1] ✓ 3s
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd

from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

print("Libraries imported")

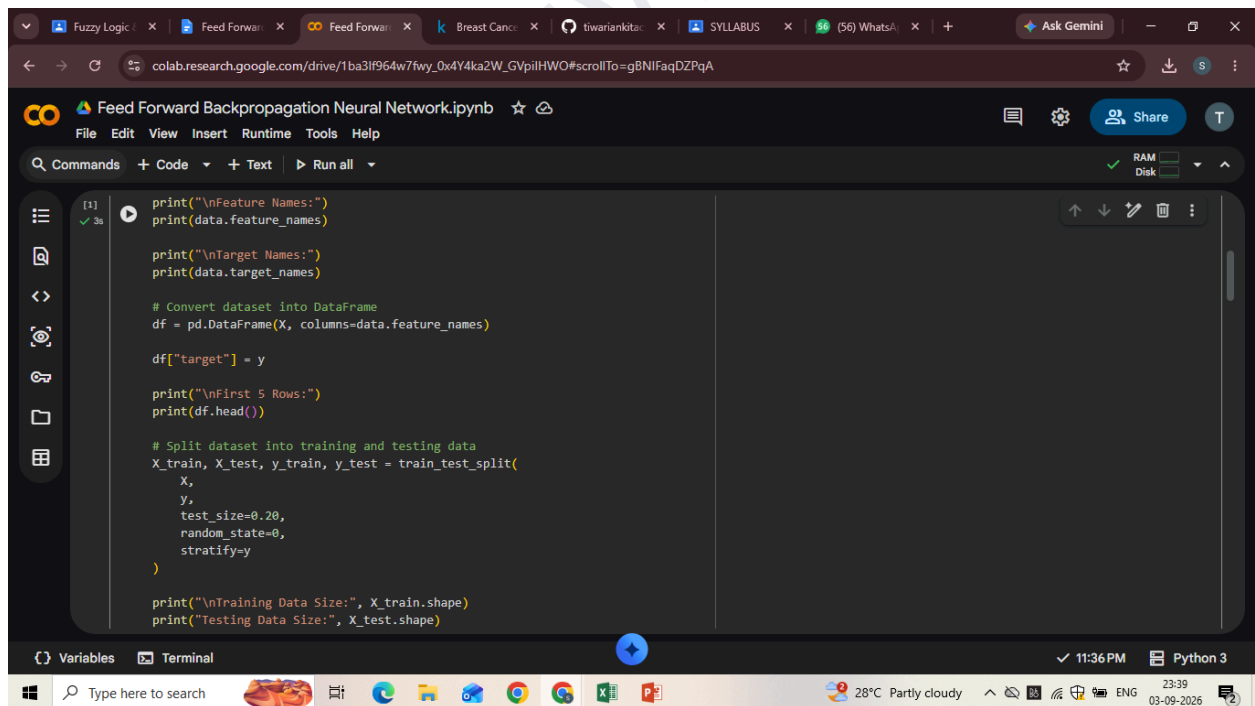
# Load Breast Cancer Dataset
data = load_breast_cancer()

X = data.data
y = data.target

print("Dataset Loaded")

print("\nDataset Shape:")
print(X.shape)

print("\nFeature Names:")
print(data.feature_names)
```



```
[1] ✓ 3s
print("\nFeature Names:")
print(data.feature_names)

print("\nTarget Names:")
print(data.target_names)

# Convert dataset into DataFrame
df = pd.DataFrame(X, columns=data.feature_names)

df["target"] = y

print("\nFirst 5 Rows:")
print(df.head())

# Split dataset into training and testing data
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=0,
    stratify=y
)

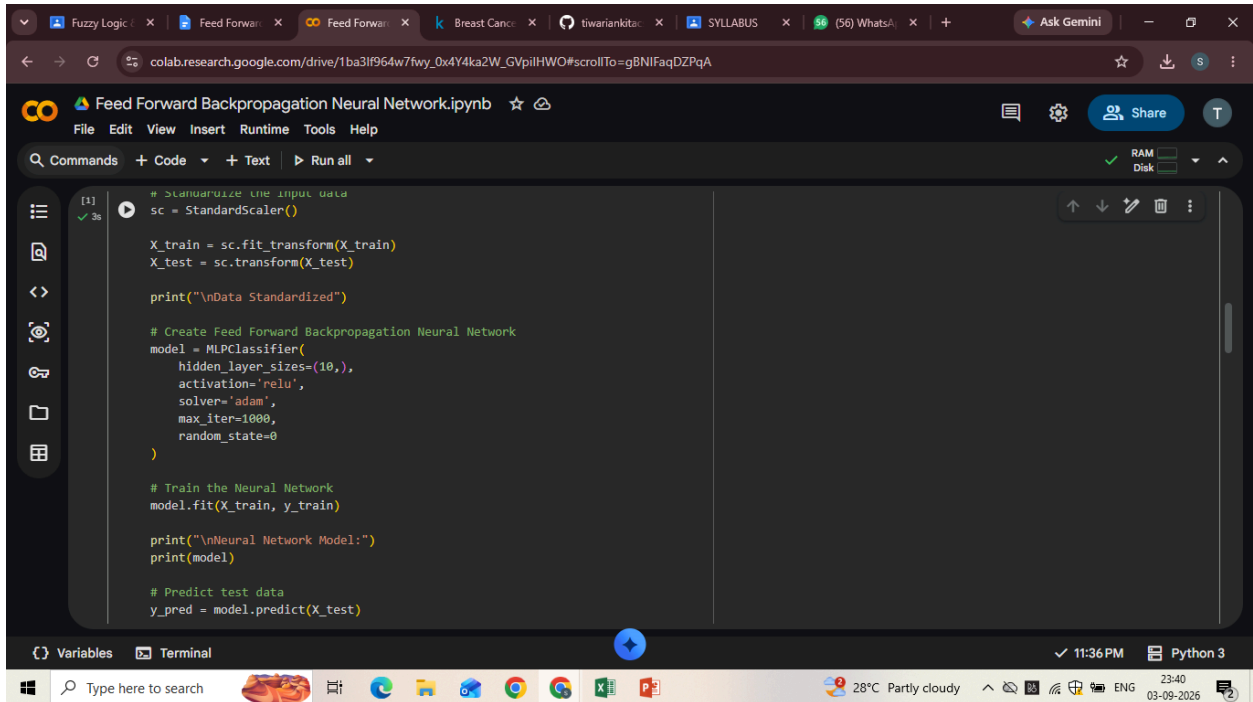
print("\nTraining Data Size:", X_train.shape)
print("Testing Data Size:", X_test.shape)
```

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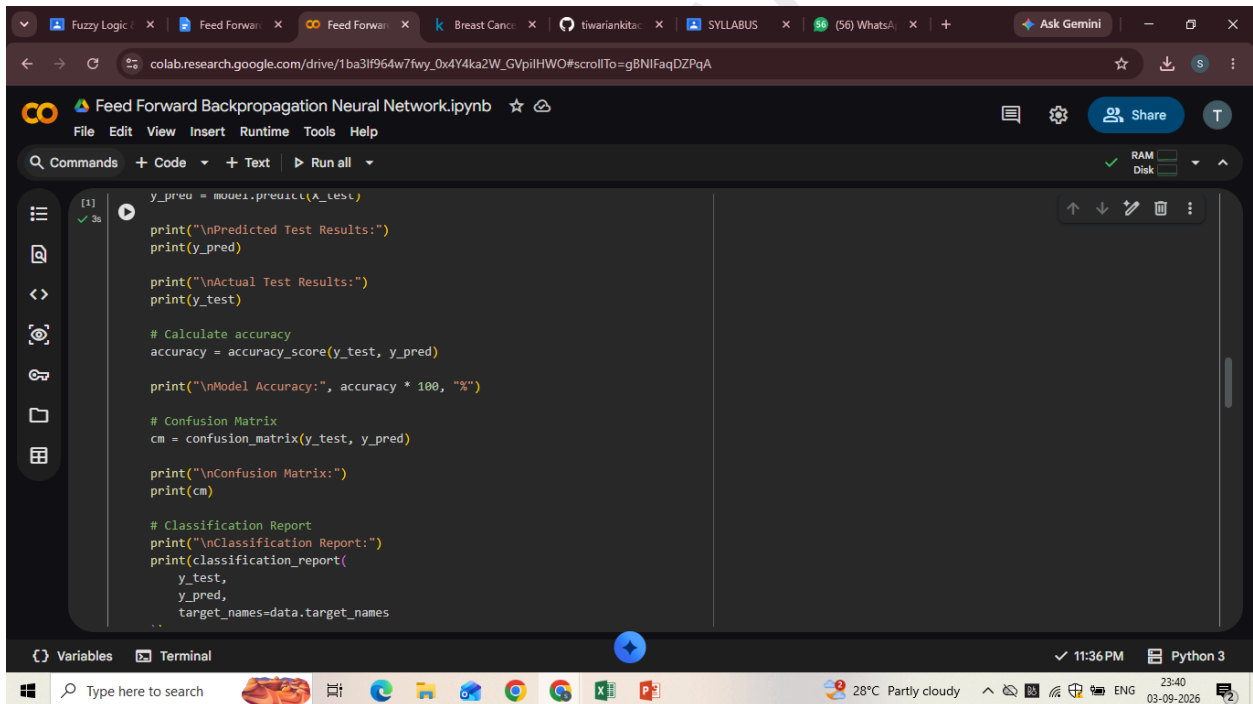
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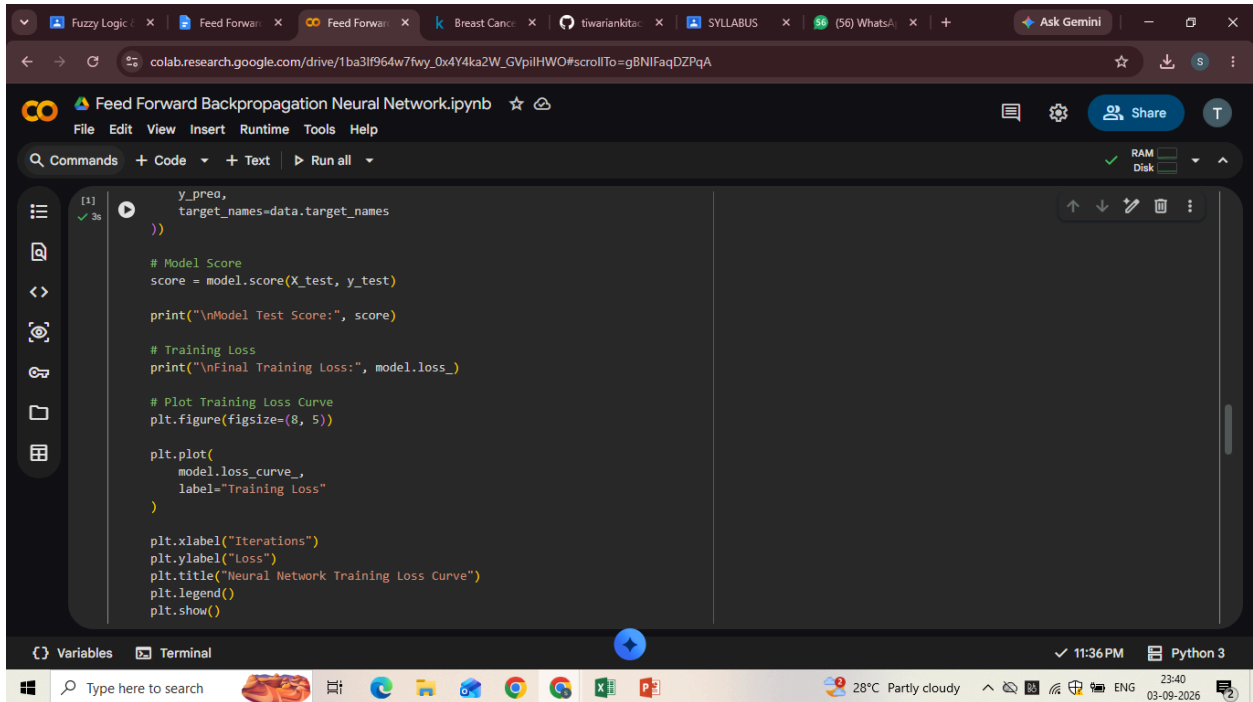
```
[1] ✓ 3s  
# Standardize the input data  
sc = StandardScaler()  
  
X_train = sc.fit_transform(X_train)  
X_test = sc.transform(X_test)  
  
print("\nData Standardized")  
  
# Create Feed Forward Backpropagation Neural Network  
model = MLPClassifier(  
    hidden_layer_sizes=(10,),  
    activation='relu',  
    solver='adam',  
    max_iter=1000,  
    random_state=0  
)  
  
# Train the Neural Network  
model.fit(X_train, y_train)  
  
print("\nNeural Network Model:")  
print(model)  
  
# Predict test data  
y_pred = model.predict(X_test)
```



```
[1] ✓ 3s  
y_pred = model.predict(X_test)  
  
print("\nPredicted Test Results:")  
print(y_pred)  
  
print("\nActual Test Results:")  
print(y_test)  
  
# Calculate accuracy  
accuracy = accuracy_score(y_test, y_pred)  
  
print("\nModel Accuracy:", accuracy * 100, "%")  
  
# Confusion Matrix  
cm = confusion_matrix(y_test, y_pred)  
  
print("\nConfusion Matrix:")  
print(cm)  
  
# Classification Report  
print("\nClassification Report:")  
print(classification_report(  
    y_test,  
    y_pred,  
    target_names=data.target_names  
))
```

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```
[1] y_pred,
    target_names=data.target_names
    ))

# Model Score
score = model.score(X_test, y_test)

print("\nModel Test Score:", score)

# Training Loss
print("\nFinal Training Loss:", model.loss_)

# Plot Training Loss Curve
plt.figure(figsize=(8, 5))

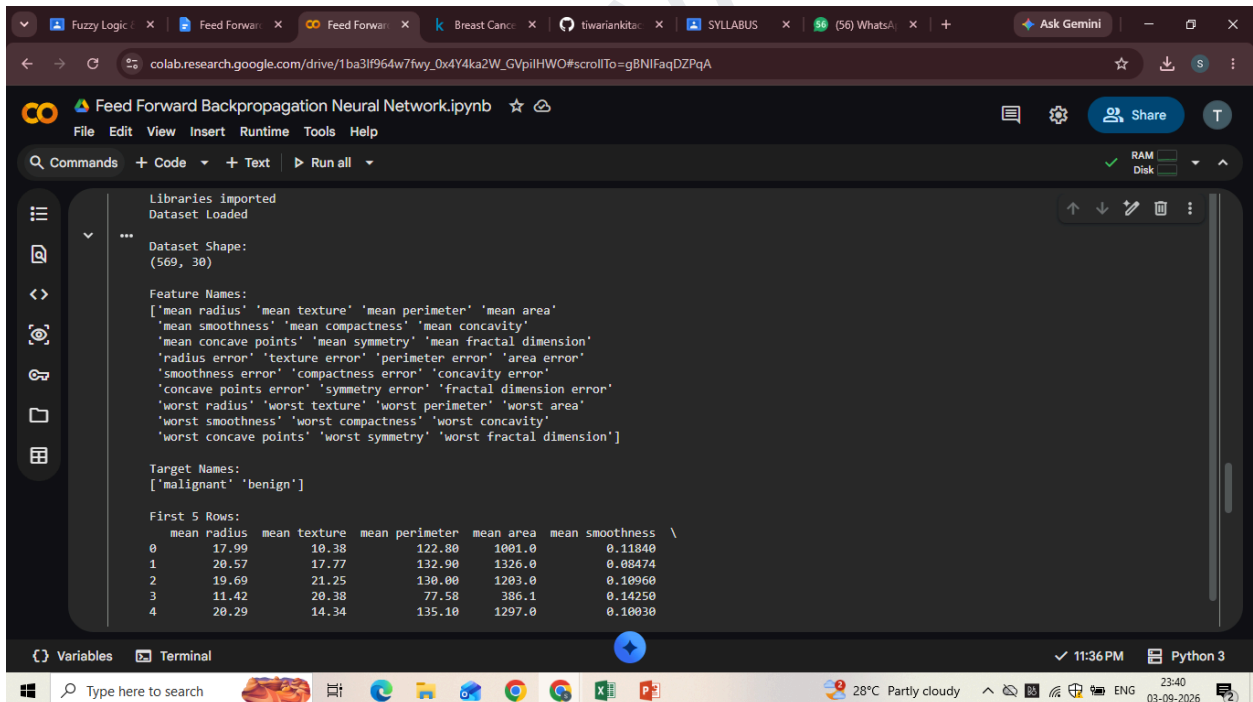
plt.plot(
    model.loss_curve_,
    label="Training Loss"
)

plt.xlabel("Iterations")
plt.ylabel("Loss")
plt.title("Neural Network Training Loss Curve")
plt.legend()
plt.show()
```

Variables Terminal

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OUTPUT:



```
Libraries imported
Dataset Loaded

Dataset Shape:
(569, 30)

Feature Names:
['mean radius' 'mean texture' 'mean perimeter' 'mean area'
 'mean smoothness' 'mean compactness' 'mean concavity'
 'mean concave points' 'mean symmetry' 'mean fractal dimension'
 'radius error' 'texture error' 'perimeter error' 'area error'
 'smoothness error' 'compactness error' 'concavity error'
 'concave points error' 'symmetry error' 'fractal dimension error'
 'worst radius' 'worst texture' 'worst perimeter' 'worst area'
 'worst smoothness' 'worst compactness' 'worst concavity'
 'worst concave points' 'worst symmetry' 'worst fractal dimension']

Target Names:
['malignant' 'benign']

First 5 Rows:
   mean radius  mean texture  mean perimeter  mean area  mean smoothness \
0         17.99         10.38         122.80      1001.0         0.11840
1         20.57         17.77         132.90      1326.0         0.08474
2         19.69         21.25         130.00      1203.0         0.10960
3         11.42         20.38          77.58       386.1         0.14250
4         20.29         14.34         135.10      1297.0         0.10030
```

Variables Terminal

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The image displays two screenshots of a Google Colab notebook titled "Feed Forward Backpropagation Neural Network.ipynb".

The top screenshot shows the input features for a breast cancer dataset, including compactness, concavity, points, symmetry, fractal dimension, texture, perimeter, area, smoothness, and targets.

	mean compactness	mean concavity	mean concave points	mean symmetry	
0	0.27760	0.3001	0.14710	0.2419	
1	0.07864	0.0869	0.07017	0.1812	
2	0.15990	0.1974	0.12790	0.2069	
3	0.28390	0.2414	0.10520	0.2597	
4	0.13280	0.1980	0.10430	0.1809	

	mean fractal dimension	...	worst texture	worst perimeter	worst area	
0	0.07871	...	17.33	184.60	2019.0	
1	0.05667	...	23.41	158.80	1956.0	
2	0.05999	...	25.53	152.50	1709.0	
3	0.09744	...	26.50	98.87	567.7	
4	0.05883	...	16.67	152.20	1575.0	

	worst smoothness	worst compactness	worst concavity	worst concave points	
0	0.1622	0.6656	0.7119	0.2654	
1	0.1238	0.1866	0.2416	0.1860	
2	0.1444	0.4245	0.4504	0.2430	
3	0.2098	0.8663	0.6869	0.2575	
4	0.1374	0.2050	0.4000	0.1625	

	worst symmetry	worst fractal dimension	target
0	0.4601	0.11890	0
1	0.2750	0.08902	0
2	0.3613	0.08758	0
3	0.6638	0.17300	0
4	0.2364	0.07678	0

The bottom screenshot shows the model training process, including data standardization, model definition (MLPClassifier), predicted and actual test results, model accuracy (96.49%), and a confusion matrix.

```
Training Data Size: (455, 30)
Testing Data Size: (114, 30)

Data Standardized

Neural Network Model:
MLPClassifier(hidden_layer_sizes=(10,), max_iter=1000, random_state=0)

Predicted Test Results:
[0 0 0 1 0 1 0 1 1 0 0 1 1 1 1 0 0 1 0 0 1 1 1 1 1 0 0 0 0 1 0 1 0 1 1 1 1
 0 1 1 0 1 0 1 1 1 1 1 1 1 0 1 0 0 1 1 1 1 1 0 0 0 1 1 1 1 1 1 1 1 1 0 1 0 1
 0 1 1 1 1 0 0 1 0 1 1 1 1 0 1 1 0 1 1 0 0 0 1 1 0 0 1 1 1 1 1 1 0 1 1 0 0
 1 1 1]

Actual Test Results:
[0 0 0 1 0 1 0 1 1 0 0 1 1 1 1 0 0 1 0 0 1 1 1 1 1 0 0 1 0 1 0 1 1 1 1
 0 1 1 0 1 1 1 1 1 1 1 1 1 0 1 0 0 0 1 1 1 1 0 0 0 1 1 1 1 1 1 1 1 0 1 0 1
 0 1 1 1 1 0 0 1 0 0 1 1 1 0 1 1 0 1 1 0 0 0 1 1 0 0 1 1 1 1 1 0 1 1 0 0
 1 1 1]

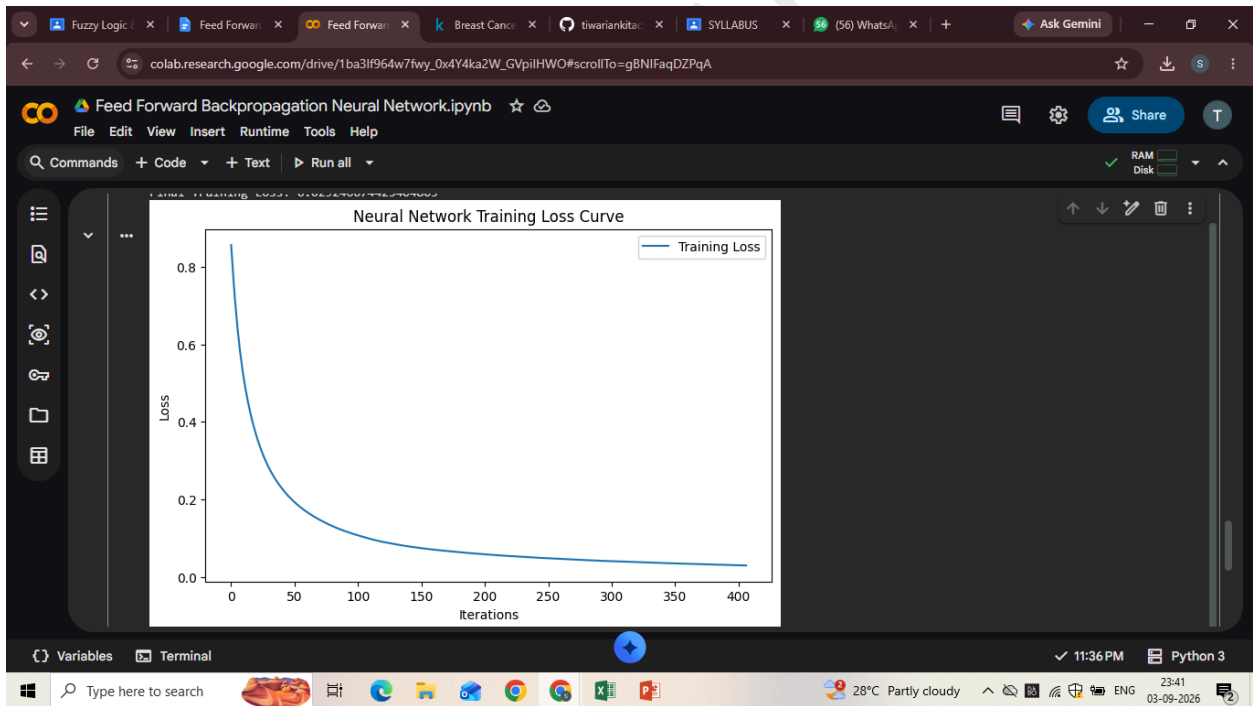
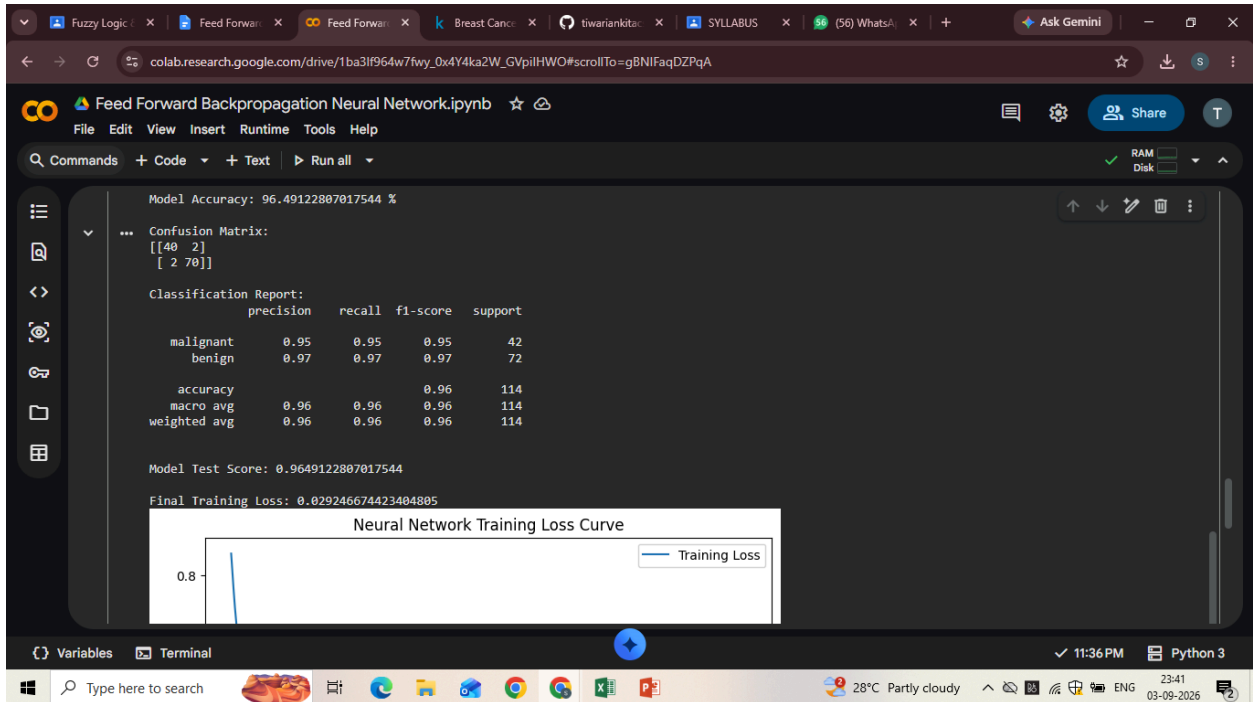
Model Accuracy: 96.49122807017544 %

Confusion Matrix:
[[40  2]
 [ 2 70]]

Classification Report:
```

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